



**Database Management
Systems(DBMS)-23AD2102A
ACADEMIC YEAR: 2024-25
Lab Workbook**

STUDENT ID:

STUDENT NAME:

Table of Contents

- 1. ER Modelling & Mapping**
- 2. Normalization Practice**
- 3. Database Creation & Management**
- 4. Basic SQL Queries**
- 5. Data Manipulation and Views**
- 6. Introduction to PL/SQL**
- 7. Implementation of Transaction in SQL Server**
- 8. Indexing Experimentation**
- 9. Query Optimization**
- 10. NoSQL Experiment**
- 11. Big Data Basics**
- 12. Replication Implementation**

Our Vision:

To be a department of international repute through continuous research, innovation and industry led curriculum.

Our Mission:

To Impart Quality Education with social consciousness and make them Globally Competent.

Organization of the STUDENT LAB WORKBOOK

The laboratory framework includes a creative element but shifts the time-intensive aspects outside of the Two-Hour closed laboratory period. Within this structure, each laboratory includes three parts: Prelab, In-lab and post-lab.

Pre-Lab

The Prelab exercise is a homework assignment that links the lecture with the laboratory period - typically takes 2 hours to complete. The goal is to synthesize the information they learn in lecture with material from their textbook to produce a working piece of software. Prelab Students attending a two-hour closed laboratory are expected to make a good-faith effort to complete the Prelab exercise before coming to the lab. Their work need not be perfect, but their effort must be real(roughly80percentcorrect).

In-Lab

The In-lab section takes place during the actual laboratory period. The First hour of the laboratory period can be used to resolve any problems the students might have experienced in completing the Prelab exercises. The intent is to give constructive feedback so that students leave the lab with working Prelab software - a significant accomplishment on their part. During the second hour, students complete the In-lab exercise to reinforce the concepts learned in the Prelab. Students leave the lab having received feedback on their Prelab and In-lab work.

Post-Lab

The last phase of each laboratory is a homework assignment that is done following the laboratory period. In the Post-lab, students analyse the efficiency or utility of a given system call. Each Post-lab exercise should take roughly 120minutes to complete.

Lab Session 01

1. ER Modelling & Mapping

Aim:

The aim of this experiment is to understand and apply the concepts of Entity Relationship (ER) Modelling to design a database schema that accurately represents the relationships between entities in a given domain.

Description:

The lab experiment on Entity Relationship (ER) Modelling involves studying and applying the fundamental principles of ER modelling to design a database schema. Students will learn how to identify and define entities, relationships, and attributes within a given domain. They will practice creating an ER diagram that accurately represents the relationships between entities and their attributes. The experiment will also cover the use of cardinality and participation constraints to establish the nature and degree of relationships. By the end of the lab, students will gain hands-on experience in translating realworld scenarios into an ER model, providing a solid foundation for database design and development. Pre-Requisites: PostgreSQL, TerraER Tool, Windows/ Ubuntu/CentOS/Debian, DBMS Concepts.

Pre-Requisites:

PostgreSQL, TerraER Tool, Windows/ Ubuntu/CentOS/Debian, DBMS Concepts.

Pre Lab-Task:

1)What is an associative entity in ER modelling, and when is it used?

2)Discuss the concept of specialization and generalization in ER modelling.

3) What is the difference between a weak entity and a strong entity in ER modelling? Provide examples of each.

4) What is the purpose of a composite attribute, and how is it represented in an ER diagram?

5) How do you handle complex relationships, such as ternary relationships, in an ER diagram?

6) Explain the concept of recursive relationships in ER modelling and provide an example.

In Lab Task:

Airline Reservation System

Problem Description: There are 6 different airlines in 6 different countries: Canada – Air Can, USA - USAir, UK – Brit Air, France – Air France, Germany – Luft Air, Italy – Ital Air. Their flights involve the following 12 cities: Toronto and Montreal in Canada, New York and Chicago in US, London and Edinburgh in UK, Paris and Nice in France, Bonn and Berlin in Germany, Rome and Naples in Italy. In each of the 12 cities, there is a (single) booking office. You are going to design a central air-reservation database to be used by all booking offices.

The flight has a unique flight number, airline code, business class indicator, smoking allowed indicator. Flight availability has flight number, date + time of departure, number of total seats available in business class, number of booked seats in business class, number of total seats available in economy class, and number of booked seats in economy class.

The customers may come from any country, not just the 6 above, and from any province/state, and from any city. The customer has first & last name, mailing address, zero or more phone numbers, zero or more fax numbers, and zero or more email addresses. Mailing address has street, city, province or state, postal code and country. Phone/fax number has country code, area code and local number. Email address has only one string, and no structure is assumed. A customer can book one or more flights. Two or more customers may have same mailing address and/or same phone number(s) and/or same fax number(s). But the email address is unique for each customer. First and last names do not have to be unique.

Booking has an unique booking number, booking city, booking date, flight number, date + time of departure (in local time, and time is always in hours and minutes), date + time of arrival (in local time), class indicator, total price (airport tax in origin + airport tax in destination + flight price – in local currency. The flight price for business class is 1.5 times of the listed flight price), status indicator (three types: booked. Canceled – the customer canceled the booking, scratched – the customer had not paid in full 30 days prior to the departure), customer who is responsible for payment, amount-paid-so far (in local currency), outstanding balance (in local currency), the first & last names to be printed on the ticket. The airport taxes must be stored in local currencies (i.e. Canadian 19 dollars, US dollars, British Pounds, French francs, German marks, and Italian Liras). Since the exchange rates change daily, they also must be stored for calculations of all prices involved. Though France, Germany, and Italy have had a common currency for a while, we used the names of their original currencies to involve in this exercise currency exchange rates and their changes.

1)Design an ER diagram that represents the entities, attributes, relationships, and cardinalities for the Airline Reservation System. Consider any necessary assumptions and justify them if required

2)Write the Relational schema that represents the entities, attributes, relationships, and cardinalities for the Airline Reservation System. Consider any necessary assumptions and justify them if required.

Viva-Voce Questions (In-Lab):

1) Can you explain the limitations or drawbacks of using an ER diagram for complex database systems?

2) How would you handle a scenario where an entity has multiple relationships with the same entity type in an ER diagram?

3) What are some common challenges or pitfalls that can arise when identifying entities and relationships in a real-world domain for ER modelling?

4) How do you handle evolving or changing requirements in ER modelling, especially when the relationships between entities need to be modified or updated?

5) Can you discuss the trade-offs between using a single ER diagram versus multiple interconnected ER diagrams for a complex database system?

Post Lab:

1) Fashion Hub Retail Company Database:

Problem Description: A retail company named "Fashion Hub" wants to design a database to manage their product inventory and sales. The company sells various fashion items such as clothing, accessories, and footwear. They have multiple physical stores located in different cities. The following information is provided to design the database:

- **Products:** Fashion Hub sells a wide range of products, including clothing, accessories, and footwear. Each product has a unique product code, name, brand, price, and quantity in stock. Additionally, each product belongs to a specific category (e.g., shirts, dresses, bags, shoes).
- **Stores:** The company operates multiple physical stores in different cities. Each store has a unique store ID, name, address, and contact information.
- **Customers:** Customers can create accounts with Fashion Hub to make purchases. Each customer has a unique customer ID, name, address, phone number, and email. The company also stores information about customer preferences and purchase history.
- **Sales:** Fashion Hub keeps track of sales transactions made by customers. Each sale has a unique sale ID, date and time, customer ID, and store ID. It also includes the total amount paid, any discounts applied, and the payment method used (e.g., cash, credit card).
- **Employees:** The company employs staff members at each store. Each employee has a unique employee ID, name, address, phone number, and position (e.g., store manager, sales associate).

Design an ER diagram that represents the entities, attributes, relationships, and cardinalities for the Fashion Hub database. Consider any necessary assumptions and justify them if required.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u>
	Marks Secured: _____ out of _____
	Full Name of the Evaluator:
	Signature of the Evaluator Date of Evaluation:

Lab Session 02

Normalization Practice

Aim:

Apply normalization techniques (1NF, 2NF, 3NF) to optimize an unnormalized database design.

Objective:

This program will implement the reference rules of the Normalization and examine how to reduce the scheme into the well-structured form. With this experiment student, can notice how the primary keys of the schema are deduced from the describe functional dependencies of the schema and determine, on which Normal Form the given schema is fitted.

Pre-Requisites:

Knowledge about Normalization and Integrity Rules, Amstrong Rules of the Functional Dependencies.

Pre-Lab Task:

1) What is the purpose of the Integrity Rules?

2) What is Normalization?

3) What are Armstrong Rules?

4) What are the differences between Integrity Rules and Normalization Process.

5) What is closure set of Functional Dependencies

In Lab Task:

1) Determine the Primary Keys for the relation $R(A,B,C,D,E)$ by using the following functional dependencies. $A \rightarrow B$, $A \rightarrow C$, $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$

2) Find out the closure set of FD's for the relation $R(A,B,C,D,E,F,G,H)$ by using following functional dependencies $AB \rightarrow CD$, $CD \rightarrow EF$, $G \rightarrow H$, $A \rightarrow H$, $ABD \rightarrow GH$, $E \rightarrow F$, $A \rightarrow H$, $DEF \rightarrow GH$

3) Find out the non-keys from the following functional dependencies for the relation $R(A,B,C,D,E)$ $A \rightarrow B$, $AB \rightarrow C$, $C \rightarrow D$, $ABC \rightarrow E$, $AB \rightarrow D$

A) Finding Non-Keys for Relation R(A, B, C, D, E):

4) Check whether the Relations R1 (A,B,C,D) and R2(A,B,C,D,E) are closer or not. The definitions and functional dependencies of R1 and R2 are as R1 FD'S: $A \rightarrow B$, $A \rightarrow C$, $A \rightarrow D$, $AB \rightarrow C$, $BC \rightarrow D$, $ABC \rightarrow D$ R2 FD'S: $A \rightarrow C$, $B \rightarrow D$, $B \rightarrow C$, $AB \rightarrow BC$, $BC \rightarrow D$, $BC \rightarrow CD$, $ABC \rightarrow ABD$

A) Checking Normal Form for Relation S(SNO, SNAME, CITY, STATUS):

5) Determine on which normal form the relation S(SNO,SNAME,CITY,STAUS) satisfies, to determine the normal form use the following functional dependencies $SNO \rightarrow CITY$, $SNO \rightarrow STATUS$, $CITY \rightarrow STATUS$, $SNO \rightarrow SNAME$

6) Deduce the well-structured relations for the Relation SHIPMENT (SNO, SNAME, CITY, STAUS, PNO, PNAME, QTY), by using following FD's SNO->CITY, SNO->SNAME, CITY->STATUS, SNO->STATUS, PNO->PNAME, (SNO, PNO)-> QTY and also find out on which normal form the deduced relation resides

Viva-Voce Questions (In-Lab):

1) What is Functional Dependency?

2) What is Transitive Dependency?

3) What are Spurious Variables?

4) What is Multi Valued Functional Dependency?

5) What is non-loss FD?

6) “A BCNF is also in 3NF”, is this statement is true and what about reversal of the statement?

Post Lab Task:

1) Prove all the Armstrong Rules of Functional Dependencies.

2) Prove all the rules of Multivalued Functional Dependencies.

3) Let us assume a table User_Personal as given below.

<i>UserID</i>	<i>U_email</i>	<i>Fname</i>	<i>Lname</i>	<i>City</i>	<i>State</i>	<i>Zip</i>
MA12	Mani@ymail.com	MANISH	JAIN	BILASPUR	CHATISGARH	458991
PO45	Pooja.g@gmail.co	POOJA	MAGG	KACCH	GUJRAT	832212
LA33	Lavle98@jj.com	LAVLEEN	DHALLA	RAIPUR	CHATISGARH	853578
CH99	Cheki9j@ih.com	CHIMAL	BEDI	TRICHY	TAMIL NADU	632011
DA74	Danu58@g.com	DANY	JAMES	TRICHY	TAMIL NADU	645018

I. Is this table in First Normal Form (1NF)?

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 03

3. Database Creation & Management

Aim/Objective:

The aim of this experiment is to familiarize students with the To download, install the PostgreSQL and practical implementation of creating database objects, defining their structure, and apply constraints using DDL operations.

Description:

PostgreSQL is a popular relational database management system (RDBMS). PostgreSQL is not controlled by any corporation or other private entity and the source code is available free of charge. PostgreSQL runs on all major operating systems, including Linux, UNIX and Windows. It supports text, images, sounds, and video, and includes programming interfaces for C / C++, Java, Perl, Python, Ruby, Tcl and Open Database Connectivity (ODBC). The "Implementation of DDL Commands" lab experiment focuses on providing students with practical exposure to executing Data Definition Language (DDL). The experiment involves creating database objects, such as tables, views, and indexes, using DDL commands. Through this hands-on experience, students will gain proficiency in effectively utilizing DDL. The lab aims to enhance students' understanding of database management and reinforce their skills in executing DDL.

Pre-Requisites:

PostgreSQL, TerraER Tool, Windows/ Ubuntu/CentOS/Debian, DBMS Concepts.

In Lab Task:

Implementation of basic PostgreSQL Queries with DDL commands.

Schema:

SAILORS (SID: INTEGER, SNAME: STRING, RATING: INTEGER, AGE: REAL);

BOATS (BID: INTEGER, BNAME: STRING, COLOR: STRING);

RESERVES (SID: INTEGER, BID: INTEGER, DAY: DATE).

SAILORS			
SID	SNAME	RATING	AGE
22	Dustin	7	45
29	Brutus	1	33
31	Lubber	8	56
32	Andy	8	26
58	Rusty	10	35
64	Horatio	7	35
71	Zor	10	16
74	Horatio	9	40
85	Art	3	26
95	Bob	3	64

BOATS		
ID	BNAME	COLOR
101	Interlake	Blue
102	Interlake	Red
103	Clipper	Green
104	Marine	Red

RESERVES		
SID	BID	DAY
22	101	1998-10-10
22	102	1998-10-10
22	103	1998-10-8
22	104	1998-10-7
31	102	1998-11-10
31	103	1998-11-6
31	104	1998-11-12
64	101	1998-9-5
64	102	1998-9-8
74	103	1998-9-8

In the reserves Questions:

- i) Create tables for Sailors, Reserves and Boats with appropriate key constraints?
- ii) Write PostgreSQL Queries to insert values as per the given above Data Base?
- iii) To retrieve distinct Sailor name and Sailor Age from Sailors?
- iv) To retrieve boat name, boat id of the red and green color boats?
- v) Find the names of sailors who have reserved a red boat, and list in the order of age?
- vi) Find the ids of sailors who have reserved a red boat or a green boat?
- vii) Find the names of sailors who have reserved at least one boat?
- viii) Display the Insert 'twenty' in age column
- ix) Display the Insert sid: 100 bid: 205, day: 12-09-17 in reserves table.
- x) Create a query that displays the Insert sid: 95, Sname: anil, Rating:3, age: 64 in sailor's table
- xi) Create a query that displays the update 10.5 as rating where sailor id= 95
- xii) table give date as per this format like did/mm/yyyy

Viva-Voce Questions (In-Lab):

1) What is PostgreSQL?

2) What are the advantages of PostgreSQL?

3) Which data types are used in PostgreSQL?

4)What are some common challenges or considerations when designing and implementing complex database schemas using DDL commands?

5)Explain the concept of data integrity constraints in DDL commands and provide examples of commonly used constraints.

Post Lab Task:

Use the tables below and implement the below queries

Customer			
CID	PID	Rental_Date	Rental_Period
104	300	10-06-2020	6
105	302	05-07-2020	10
108	304	15-07-2020	3
109	305	25-06-2020	6

Artist			
Aid	Name	Address	Phone
200	John	Delhi	7786549803
201	Samuel	Mumbai	7123458790
202	Samson	Lucknow	9460367777
203	David	Hyderabad	9797276764
204	Raghu	Hyderabad	8134185751
205	Ravi	Mumbai	7471094738
206	Kiran	Delhi	9808003725

Rent			
Pid	Aid	Rental_Cost	Type
300	201	4500	Hired
301	202	3500	Not Hired
302	203	7500	Hired
303	205	2500	Not Hired
304	202	10000	Not Hired
305	201	8000	Not Hired
306	203	6500	Not Hired

Painting				
Cid	Cname	Address	Phone	Category
100	Raju	Hyderabad	9876045789	Bronze
101	Hari	Vijayawada	8877678956	Gold
102	Devi	Guntur	7879312123	Silver
103	Rani	Delhi	8780945290	Platinum
104	Jaya	Mumbai	9612578457	Gold
105	Haritha	Kolkata	9611665513	Silver
106	Kalyan	Vijayawada	9610752569	Bronze
107	Roja	Hyderabad	9609839625	Platinum
108	Amar	Vijayawada	9608926681	Gold
109	Padma	Vijayawada	9608013737	Bronze

Owner				
Pid	Oid	Name	Address	Phone
300	500	Raju	Hyderabad	9460367777
301	500	Hari	Vijayawada	8134185751
302	501	Giri	Hyderabad	7808003725
303	501	Gopi	Delhi	9481821699
304	503	Krishna	Mumbai	7155639673
305	502	Verma	Delhi	8829457647
306	502	Guna	Delhi	7503275621

Create the tables identifying the constraints and relationships.

i) Insert atleast 10 records into these tables

ii) Display the customer details who got the category as “Gold”

iii) Display the list of artists who belong to “Hyderabad”

iv) Create a PostgreSQL query display the painting details which are not hired

v) Create a PostgreSQL query to update the rental cost of the painting with 1000 for the paintings which are not hired.

vi) Display the details of the artist whose paintings are hired

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 04

4. Basic SQL Queries

Aim:

The aim of this experiment is to familiarize students with the syntax and usage of these join operations, enabling them to effectively combine data from multiple tables based on specified conditions.

Description:

The "Implementation of Joins: Inner Join, Outer Join, Natural Join" lab experiment involves a procedural description where students gain hands-on experience in executing different types of joins in a database system. The lab begins with a brief introduction to the concepts of joins, emphasizing Inner Join, Outer Join, and Natural Join. Students then proceed to practice implementing these join operations using SQL queries. They learn how to write queries that combine data from multiple tables based on specified conditions, such as matching values in related columns. The lab provides students with sample datasets and real-world scenarios, enabling them to apply the appropriate join type to retrieve the desired results. Through this procedural description, students enhance their understanding of join operations and develop the skills needed to effectively query and analyze relational databases using Inner Join, Outer Join, and Natural Join.

Pre-Requisites:

PostgreSQL, TerraER Tool, Windows/ Ubuntu/CentOS/Debian, DBMS Concepts

Pre Lab-Task:

1) What Explain the concept of Inner Join in PostgreSQL. How does it work, and what is its purpose?

2) What are the key differences between an Inner Join and a Cross Join in PostgreSQL?

3) Explain the types of Outer Joins supported by PostgreSQL: Left Outer Join, Right Outer Join, and Full Outer Join.

4) When would you use a Natural Join in PostgreSQL? Give an example scenario where a Natural Join is appropriate.

5) What are some best practices when working with Joins in PostgreSQL to optimize performance and avoid common pitfalls?

In Lab Task:

1. Implementation of distinct types of Joins: Inner Join, Outer Join, Natural Join etc.

Entities	Attributes
Customer	Cust_Id, Cust_Name, Dob, City, Street, State, Pincode, Ph_No, Deal_No, Photo_Identity, V_Id
Vehicle	V_Id, Veh_Type, Veh_Name, Veh_Number
Edu_Bus	Edu_Id, Edu_Name, Ph_No, City, Street, State, Pincode, Deal_No
Dealer	Deal_Id, Deal_Name, City, Street, State, Pincode, D_No, Ph_Int
Branch	Branch_Id, B_Name, State, City, Pincode, Street, D_No, Phno1, Phno2, C_Id, V_Id, E_Id
Renewal	Brach_Id, Cid, Check_License_Period
Registration	Cust_Id, V_Id, Dealid, Date
Contract_Permission	V_Id, Branch_Id, No_Of_Days, Amount_Per_Seat

CUSTOMER

cust_id	cust_name	dob	city	street	state	pincode	ph_no	deal_no	photo_identity	v_id
41	raju	13-09-1996	Guntur	Ramgopal	Andhra_pradesh	5000213	9123456789	10	y	3
42	heri	19-06-2016	Perambur	Mylapur	Tamil_Nadu	500211	1122334455	20	n	2
43	giri	20-01-1995	hyderabad	srnagar	Telangana	500079	8877665544	30	y	4
44	ramu	17-07-1996	vijayawada	benz circle	Andhra_pradesh	512345	7654564321	40	y	5
45	rahu	08-12-1995	guntur	rajunagar	Andhra_pradesh	523022	9999999998	50	y	7
46	gopi	13-08-1979	hyderabad	gachibowli	Telangana	567089	7787777775	10	n	1
47	karthik	15-01-2004	guntur	chandram	Andhra_pradesh	546789	7788776633	20	n	6
48	gopal	06-12-2000	Hyderabad	ameerpet	Telangana	500023	6734556345	30	y	8
49	dinesh	10-12-2001	Hyderabad	kondapur	Telangana	502033	6794537212	30	n	10
50	suresh	25-03-2025	vijayawada	poranki		512022	7896543233	20	y	9

VEHICLE

veh_id	veh_type	veh_name	veh_number
1	2_wheeler	royal_enfield	AP1234
2	3_wheeler	auto	AP3421
3	2_wheeler	royal_enfield	TS213
4	4_wheeler	fiat	AP2346
5	4_wheeler	benz	TS1256
6	3_wheeler	auto	TN5544
7	2_wheeler	splendor	AP3214
8	2_wheeler	bojaj	AP7895
9	2_wheeler	royal_enfield	AP2134
10	4_wheeler	ambassador	TS4567

EDU_BUS

edu_id	edu_name	ph_no	city	street	state	pincode	deal_no
31	dps	1122334455	Hyderbad	santhnagar	Telangana	512345	444
32	klu	44556677	guntur	vaddeswaram	Andhra pradesh	567432	111
33	dav	123456789	Hyderbad	jubilee hills	Telangana	500897	333
34	surya	4356789321	Hyderbad	bachupally	Telangana	512098	111
35	vit	7788996578	Hyderbad	kukatpally	Telangana	523087	222
36	rvrrjc	2233445566	Guntur	guntur	Andhra pradesh	512087	222
37	vnr	1122334455	Hyderbad	miyapur	Telangana	512345	333
38	klh	3445996578	Hyderbad	aziznagar	Telangana	512345	222
39	bvrit	112566725	Hyderbad	nizampet	Telangana	512345	111
40	cbit	1122965785	Hyderbad	gandipet	Telangana	512345	111

DEALER

deal_id	deal_name	city	street	state	pincode	d_no	ph_int
51	raju	guntur	Raju Nagar	Andhra Pradesh	612345	555	9988776655
52	raghu	hyderabad	kukatpally	Telangana	678890	666	8765489765
53	kiran	hyderabad	bachupally	Telangana	546789	777	7654564556
54	ganesh	hyderabad	kondapur	Telangana	456789	111	874648545
55	hari	hyderabad	ammerpet	Telangana	534467	222	9988776655
56	kiran	hyderabad	santhanagar	Telangana	512334	333	9988776655
57	kamal	hyderabad	miyapur	Telangana	504406	444	9988776655
58	eswar	guntur	mangalagiri	Andhra Pradesh	563456	888	9988776655

BRANCH

branch_id	b_name	state	city	pincode	street	d_no	phno1	phno2	c_id	v_id	e_id
210	gandipet	Telangana	Hyderabad	512345	jntu	53	9848022338	8802233811	41	1	31
211	madhapur	Telangana	Hyderabad	512345	gachibowli	52	9848022344	8802234433	43	2	32
212	vaddeswaram	Andhra pradesh	Guntur	567432	kondapur	51	8802233811	9848022311	41	1	31
213	jubilee hills	Telangana	Hyderabad	500897	ameerpet	53	8802234433	9848065722	43	2	32
214	bachupally	Telangana	Hyderabad	512098	erragada	55	9848022311	8065722322	41	1	31
215	kukatpally	Telangana	Hyderabad	523087	prbs	52	9848065722	8065722311	43	2	32
216	guntur	Andhra pradesh	Guntur	512087	towers	54	8880557222	9848022338	41	1	31
217	miyapur	Telangana	Hyderabad	512345	bhaskar nagar	55	8806572233	9848022338	41	1	31
218	aziznagar	Telangana	Hyderabad	512345	rajunagar	56	8065722322	9848022344	43	2	32
219	nizampet	Telangana	Hyderabad	512345	kbhp	52	8065722311	9848022338	41	1	31

RENEWAL

branch_id	c_id	check_license_period
210	41	4
210	42	6
213	44	4
211	45	9
211	46	10
215	47	4
216	48	6
217	49	7
217	50	8

REGISTRATION

cust_id	veh_id	deal_id	date
41	3	55	04-04-2014
42	2	54	02-09-2016
43	4	55	03-12-2015
44	5	52	29-09-2016
45	7	55	18-11-2013
46	1	51	06-10-2014
47	6	52	11-07-2011
48	8	52	12-06-2015
49	10	53	02-03-2014
50	9	53	20145-10-11

CONTRACT_PERMISSION

veh_id	branch_id	no_of_days	amount_per_seat
4	210	15	200
5	210	43	100
10	212	15	400

Questions:

- i) Create the database in PostgreSQL and create the necessary tables for the given case study using appropriate keys and relationships between the tables**
- ii) Display the vehicles that were registered by the dealer name 'Raghu'**
- iii) Display the list of customers who have applied for new license**
- iv) Display the vehicles who have been given 30 days of contract permission.**
- v) Create a query to display all the records who applied for renewal of license.**
- vi) Display the count of vehicles of different types.**
- vii) Create a query to display customer details who have 2-wheeler vehicle.**
- viii) Create a query that displays the customer details whose license expires in 5 days.**
- ix) Display the list of educational institutions who applied for permit**
- x) Display the total number of vehicles license allotted by each branch**
- xi) Display the number of customer present under each dealer.**

2. Use the tables below and implement the below queries

Student:

REGNO	NAME	ADDRESS	PHONE	AGE
101	Sai	Vijayawada	9455123451	18
102	Teja	Guntur	9652431543	18
103	Dinesh	Delhi	9156253131	20
104	Taurn	Chennai	9156768971	18
105	Dhanraj	Mumbai	9156253132	19
106	Rohit	Bangalore	9652431544	18
107	Niraj	Goa	9455123452	18

Course:

COURSE_ID	REG NO
1	101
2	102
2	103
3	104
4	105
5	106
1	107

- i) Write a PostgreSQL query to retrieve the details of students who have enrolled in Course 1
- ii) How can you obtain the names of students who have enrolled in Course 2 using an Inner Join between the Student and Course tables?
- iii) Write a PostgreSQL query to fetch the names and phone numbers of students who have not enrolled in any course.
- iv) How can you retrieve the list of courses along with the corresponding student names using an Inner Join between the Student and Course tables?
- v) Write a PostgreSQL query to find the number of students enrolled in each course. Include the course name and the count of students.
- vi) How can you retrieve the names of students along with their corresponding course IDs and course names, even if they have not enrolled in any course? Use an Outer Join for this query.
- vii) Write a PostgreSQL query to obtain the names of students who are enrolled in Course 1 and Course 2 simultaneously.

viii) How can you retrieve the names of students along with their course details, sorted by their age in descending order? Use appropriate join and sorting techniques.

ix) Write a PostgreSQL query to find the students who are enrolled in courses but have an age greater than 18.

x) How can you retrieve the details of students who have the same course enrollment as the student with REGNO 101?

Viva-Voce Questions (In-Lab):

1) What is the purpose of using joins in a database query, and how do they help retrieve meaningful information from multiple tables?

2) When would you choose to use an Inner Join instead of an Outer Join, and vice versa?

3) Can you provide an example of a real-world scenario where an Inner Join is more appropriate than an Outer Join?

4) How can you handle situations where duplicate column names exist in the joined tables during a Natural Join?

5) What are the benefits and limitations of using the USING clause in a join operation?

Post Lab Task:

1) Use the tables below and implement the below queries

Employee:

EMPID	EMPFNAME	EMPLNAME	AGE	EMAILID	PHONENO	ADDRESS
1	Vardhan	Kumar	22	vardy@abc.com	9876543210	Delhi
2	Himani	Sharma	32	himani@abc.com	9977554422	Mumbai
3	Aayushi	Shreshth	24	aayushi@abc.com	9977555121	Kolkata
4	Hemanth	Sharma	25	hemanth@abc.com	9876545666	Bengaluru
5	Swatee	Kapoor	26	swatee@abc.com	9544567777	Hyderabad

Project:

PROJECTID	EMPID	CLIENTID	PROJECTNAME	PROJECTSTARTDATE
111	1	3	Project1	2019-04-21
222	2	1	Project2	2019-02-12
333	3	5	Project3	2019-01-10
444	3	2	Project4	2019-04-16
555	5	4	Project5	2019-05-23
666	9	1	Project6	2019-01-12
777	7	2	Project7	2019-07-25
888	8	3	Project8	2019-08-20

Questions:

I) Write a PostgreSQL query to retrieve the details of employees who are assigned to projects

II) How can you obtain the names and project names of employees using an Inner Join between the employee and Project tables?

III) Write a PostgreSQL query to fetch the details of employees who have not been assigned to any project.

IV) How can you retrieve the names and email IDs of employees along with their corresponding project names, even if they have not been assigned to any project? Use an Outer Join for this query.

V) Write a PostgreSQL query to find the number of projects assigned to each employee. Include the employee name and the count of projects.

VI) How can you retrieve the names of employees along with their corresponding project details, sorted by their age in ascending order? Use appropriate join and sorting techniques

VII) Write a PostgreSQL query to obtain the names of employees who are assigned to both Project 1 and Project 2 simultaneously

VIII) How can you retrieve the names of employees along with their project details, sorted by the project start date in descending order? Use appropriate join and sorting techniques.

IX) Write a PostgreSQL query to find the employees who are assigned to projects but have an age greater than 25.

X) How can you retrieve the details of employees who have the same project assignment as the employee with EMPID 3?

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u> 	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 05

5. Data Manipulation and Views

Aim:

The aim of this experiment is to familiarize students with the concept of DML commands, subqueries, their syntax, and usage for retrieving specific data subsets within queries. Additionally, the lab focuses on DML commands and creating views, which are virtual tables derived from queries, to simplify complex queries and enhance data accessibility.

Description:

The lab experiment involves a procedural description where students gain hands-on experience in implementing DML commands, subqueries and creating views in a database system. The lab begins with an introduction to the DML commands like INSERT, UPDATE, DELETE, subqueries, explaining their purpose and syntax. Students then proceed to practice writing queries with subqueries, embedding them within SELECT, FROM, and WHERE clauses to retrieve specific subsets of data based on conditions. They learn how to use subqueries to perform calculations, filtering, and data manipulation. The lab also covers the creation of views, allowing students to create virtual tables derived from queries to simplify complex queries and improve data accessibility. Students learn to create, modify, and utilize views in their queries.

Pre Lab-Task:

1)How can you delete rows from a table.

2) How to Update Rows in a Predefined Table

3) What is a View in PostgreSQL, and What Purpose Does it Serve?

4)How to Create a View in PostgreSQL using the CREATE VIEW statement?

5)Can You Modify Data Through a View in PostgreSQL? Why or Why Not?

6)How to Update the Definition of a View in PostgreSQL Using the ALTER VIEW Statement

In Lab Task:

1) Write PostgreSQL queries using DML commands and nested subqueries to analyze the following data.

Patient:

Pid	Fname	Lname	Email	Phno	Address	Date_Admn
200	David	Samson	aaa@gmail.com	7654328976	Mumbai	13-10-2019
201	Bharath	Kumar	bbb@gmail.com	8765438907	Hyderabad	22-01-2020
202	Eswar	Prasad	ccc@gmail.com	9876548838	Vijayawada	15-03-2020
203	Jaya	Laxmi	ddd@gmail.com	8765489765	Delhi	06-07-2020
204	Laxmi	Devi	eee@gmail.com	7654430692	Mumbai	12-03-2020
205	Pramod	Reddy	fff@gmail.com	6543371619	Hyderabad	05-02-2020
206	Charan	Kumar	ggg@gmail.com	5432312546	Vijayawada	22-04-2020
207	Kalyan	Reddy	hhh@gmail.com	4321253473	Delhi	10-07-2020
208	Rajesh	Yadav	iklmn@gmail.com	3210194400	Chennai	03-03-2020
209	Naveen	Kumar	jghfr@gmail.com	2099135327	Hyderabad	22-04-2020

Pcase	
Bednum	Pid
20	200
21	201
22	202
23	203
24	204
25	205
26	206
27	207
28	208
29	209

Shift			
Doctor Id	Day	Starttime	Endtime
100	12-04-2020	9	5
101	20-06-2020	4	8
102	01-08-2020	6	12
105	05-08-2020	13	18
106	13-10-2019	4	8
107	22-01-2020	6	12
108	15-03-2020	13	18
109	06-07-2020	9	5
110	12-03-2020	4	8
111	05-02-2020	6	12
112	22-04-2020	13	18
113	10-07-2020	4	8
114	03-03-2020	6	12
115	22-04-2020	13	18

Appointment	
Pid	Amt
200	200
201	200
202	300
203	600
204	200
205	300
206	200
207	200
208	200
209	600

Questions:

i) Write a query to insert multiple rows in a appointment table as shown above.

ii) Write a query to update all the rows of personal emails into academic mail addresses and display a patient table.

iii) Write query for creating view from patient table.

iv) Write a PostgreSQL query to update view the Patient view where patientid >3 by using replace view statement.

v) Display the patient date of joining into format “dd month yyyy” for example as “24 March 2020”.

vi) Write a PostgreSQL query to display the count of patients who come from the same city.

vii) Write a PostgreSQL query to first name of patient who admitted and allotted the bed no.

viii) Write a query to retrieve the first name and last name of patients who have an appointment, and the appointment amount is greater than the average appointment amount.

ix) Create a view that shows the number of appointments made by each doctor, along with their total earnings from those appointments.

OUTPUT:

x) Write a query to find the total number of patients admitted on each date.

xi) Create a view that displays the doctor ID, day, and total number of appointments scheduled for each day.

xii) Write a query to find the patients who have not made any appointments.

xiii) Create a view that shows the number of patients admitted on each date and the corresponding doctor ID who treated them.

xiv) Write a query to calculate the average appointment amount for each doctor, along with the doctor's first name and last name.

xv) Create a view that displays the details of patients who have been admitted to a particular city more than once.

Viva-Voce Questions (In-Lab):

1)List various DML commands.

2)Differentiate the commands ALTER and UPDATE.

3) How can you optimize performance when working with subqueries in PostgreSQL?

5) What are the benefits of using views in a database system?

6) Can you provide an example of a situation where creating a view would be beneficial?

7) Discuss the potential limitations or considerations when using subqueries or views in complex queries.

Post Lab Task:

1) Write PostgreSQL queries using nested subqueries to analyze the following data.

Questions:

i) Write a query to find the top 3 doctors who have earned the highest total amount from appointments.

ii) Create a function that takes a doctor ID as input and returns the total number of appointments made by that doctor.

iii) Write a query to find the patients who have spent the most on appointments, including their total amount spent and the doctor they visited the most.

iv) Create a view that displays the average appointment amount for each doctor, categorized by the year of the appointment.

v) Write a query to find the doctor who has treated the highest number of unique patients.

vi) Create a function that calculates the total earnings of a doctor, given their doctor ID and a date range.

vii) Write a query to find the patients who have made appointments with doctors from multiple cities.

viii) Create a view that's hows the top 5 cities with the highest number of appointments.

ix) Write a query to find the doctor who has the highest average appointment amount.

x) Create a function that takes a patient ID as input and returns the number of appointments made by that patient within a specific year.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 06

6. Introduction to PL/SQL

Aim:

The aim of this experiment is to familiarize students with the PL/SQL, programs, creating reusable procedures & functions and its capabilities for developing database-centric applications.

Description:

The lab experiment involves a procedural description where students gain hands-on experience in writing and executing PL/SQL programs, procedures, and functions. The lab begins with a brief introduction to the PL/SQL language, highlighting its role in developing database-centric applications. Students then proceed to practice implementing PL/SQL programs by writing code blocks that include variable declarations, control structures, and SQL statements. They learn how to create reusable procedures and functions to encapsulate specific logic or calculations. The lab provides students with sample scenarios and real-world problems, enabling them to apply PL/SQL programming concepts to solve database-related tasks. Through this procedural description, students enhance their understanding of PL/SQL programming, develop skills in writing efficient and maintainable code, and gain insights into the benefits of using PL/SQL for database development.

Pre Lab-Task:

1) What Compare SQL & PL/SQL

2) How do you declare and initialize variables in a PL/SQL program or procedure in PostgreSQL?

3) Can a PL/SQL function return multiple values in PostgreSQL? If so, how can it be achieved?

4) What are the benefits of using packages in PL/SQL programming in PostgreSQL? Provide an example of a situation where packages can be useful.

5) How can you handle transactions within PL/SQL programs or procedures in PostgreSQL? Explain the use of the COMMIT and ROLLBACK statements.

6) What is the difference between an anonymous PL/SQL block and a stored procedure in PostgreSQL? When would you choose to use one over the other?

7) How can you debug PL/SQL code in PostgreSQL? Describe some debugging techniques or tools available.

8) Can you call a PL/SQL function from a SQL statement in PostgreSQL? If so, provide an example and explain the process

In Lab Task:

1. Write Use the tables below and implement the below queries

Student:

REGNO	NAME	ADDRESS	PHONE	AGE
101	Sai	Vijayawada	9455123451	18
102	Teja	Guntur	9652431543	18
103	Dinesh	Delhi	9156253131	20
104	Taurn	Chennai	9156768971	18
105	Dhanraj	Mumbai	9156253132	19
106	Rohit	Bangalore	9652431544	18
107	Niraj	Goa	9455123452	18

Course:

COURSE ID	REG NO
1	101
2	102
2	103
3	104
4	105
5	106
1	107

Execute the following queries:

i) Write a PL/SQL program that retrieves all the records from the "Student" table and displays the student details.

ii) Create a PL/SQL procedure that takes a student's registration number as input and updates their age to 20. Test the procedure by updating the age of a specific student.

iii) Develop a PL/SQL function that calculates the average age of all students in the "Student" table and returns the result.

iv) Create a PL/SQL procedure that inserts a new row into the "Course" table, enrolling a student with a given registration number into a specified course. Test the procedure by enrolling a student in a course.

v) Write a PL/SQL function that retrieves the count of students in the "Student" table and returns the result.

vi) Develop a PL/SQL procedure that deletes a student's record from the "Student" table based on their registration number. Test the procedure by deleting a specific student's record.

vii) Write a PL/SQL program that retrieves all the records from the "Course" table and displays the course details.

viii) Create a PL/SQL function that counts the number of students enrolled in each course. and returns the result as a cursor.

ix) Develop a PL/SQL procedure that updates the address of a student based on their registration number. Test the procedure by updating a specific student's address.

x) Write a PL/SQL program that accepts an age as input and retrieves all the student details from the "Student" table with that age

Viva-Voce Questions (In-Lab):

1. How can you pass parameters to PL/SQL procedures and functions, and what are the benefits of doing so?

2. What are the benefits of encapsulating logic within procedures and functions in PL/SQL?

3. Discuss the usage of cursors in PL/SQL and their significance in handling result sets.

4. Explain the concept of packages in PL/SQL and how they aid in organizing and modularizing code.

5. How can you use PL/SQL programs, procedures, and functions to enhance the performance and efficiency of database operations?

Post Lab Task:

1. Consider the following Relational Table as shown in below.

Patient:

Pid	Fname	Lname	Email	Phno	Address	Date_Admn
200	David	Samson	aaa@gmail.com	7654328976	Mumbai	13-10-2019
201	Bharath	Kumar	bbb@gmail.com	8765438907	Hyderabad	22-01-2020
202	Eswar	Prasad	ccc@gmail.com	9876548838	Vijayawada	15-03-2020
203	Jaya	Laxmi	ddd@gmail.com	8765489765	Delhi	06-07-2020
204	Laxmi	Devi	eee@gmail.com	7654430692	Mumbai	12-03-2020
205	Pramod	Reddy	fff@gmail.com	6543371619	Hyderabad	05-02-2020
206	Charan	Kumar	ggg@gmail.com	5432312546	Vijayawada	22-04-2020
207	Kalyan	Reddy	hhh@gmail.com	4321253473	Delhi	10-07-2020
208	Rajesh	Yadav	iklmn@gmail.com	3210194400	Chennai	03-03-2020
209	Naveen	Kumar	jghfr@gmail.com	2099135327	Hyderabad	22-04-2020

Bednum	Pid
20	200
21	201
22	202
23	203
24	204
25	205
26	206
27	207
28	208
29	209

Shift			
Doctor Id	Day	Starttime	Endtime
100	12-04-2020	9	5
101	20-06-2020	4	8
102	01-08-2020	6	12
105	05-08-2020	13	18
106	13-10-2019	4	8
107	22-01-2020	6	12
108	15-03-2020	13	18
109	06-07-2020	9	5
110	12-03-2020	4	8
111	05-02-2020	6	12
112	22-04-2020	13	18
113	10-07-2020	4	8
114	03-03-2020	6	12
115	22-04-2020	13	18

Appointment	
Pid	Amt
200	200
201	200
202	300
203	600
204	200
205	300
206	200
207	200
208	200
209	600

Questions:

- i) Write a PL/SQL program that retrieves all the records from the "PATIENT" table and displays the patient details.**
- ii) Create a PL/SQL procedure that takes a patient's ID as input and updates their phone number in the "PATIENT" table. Test the procedure by updating the phone number of a specific patient.**
- iii) Develop a PL/SQL function that calculates the average appointment amount from the "APPOINTMENT" table and returns the result.**
- iv) Create a PL/SQL procedure that inserts a new patient record into the "PATIENT" table with the provided details. Test the procedure by adding a new patient record.**
- v) Write a PL/SQL function that retrieves the count of patients from the "PATIENT" table and returns the result.**
- vi) Develop a PL/SQL procedure that deletes a patient's record from the "PATIENT" table based on their ID. Test the procedure by deleting a specific patient's record.**
- vii) Write a PL/SQL program that retrieves all the records from the "SHIFT" table and displays the doctor shift details.**
- viii) Create a PL/SQL function that counts the number of patients per bed from the "PCASE" table and returns the result as a cursor.**
- ix) Develop a PL/SQL procedure that updates the end time of a doctor's shift based on their ID and the date. Test the procedure by updating a specific doctor's shift end time.**
- x) Write a PL/SQL program that accepts an address as input and retrieves all the patient details from the "PATIENT" table with that address.**

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u> 	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 07

7. Implementation of Transaction in SQL Server

Pre Lab-Task:

1) What are the Transactions?

2) Explain ACID properties?

3) List the transaction states.

4) Difference between commit and roll back operations.

In Lab Task:

1) Create user and implement the following commands on relation (Emp and Dept)

2) Delete those records from the table which have age = 20 and then ROLLBACK the changes in the database.

3) Update those records from the table which have age = 30 and then ROLLBACK the changes in the database.

4) Insert the records from the table Employees ('Tom Harris', 28, 'Sales'), ('Sara Parker', 22, 'HR').

5) Delete those records from the table which have age = 25 and then COMMIT the changes and then try to use ROLLBACK to the previous save point. What have you observed at this stage?

6) Develop a query to grant all privileges of employees table into departments table.

7) Develop a query to grant some privileges of employees table into departments table.

8) Develop a query to revoke all privileges of employees table from departments table.

9) Develop a query to revoke some privileges of employees table from departments table.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	--

Lab Session 08

8. Indexing Experimentation

Aim:

To evaluate the impact of various index types on query performance in a dataset, and to analyze the effectiveness of indexes in optimizing data retrieval operations.

Description:

Indexes are essential in database management systems to expedite data retrieval operations. Different index types (e.g., B-tree, Hash, Bitmap) have distinct characteristics that can significantly impact query performance. In this experiment, we aim to create multiple index types on a dataset, measure query performance, and analyze the influence of indexes on query execution time.

Prerequisites:

1. Understanding of database management systems and SQL.
2. Familiarity with index structures and their functionalities.
3. Basic knowledge of query optimization techniques.

Pre-Lab:

1. Define what an index is in the context of a database.

2. Why are indexes crucial for efficient data retrieval?

3. Explain the differences between B-tree, Hash, and Bitmap indexes. What are the strengths and weaknesses of each type?
4. How does the choice of index type affect query performance in a database?
5. Discuss the factors that influence the selection of an appropriate index type for a given dataset.

In Lab Task:

1. Select a suitable dataset for experimentation. Here we taken Orders dataset as shown below.

<u>order_id</u>	<u>customer_id</u>	<u>product_id</u>	<u>order_date</u>	<u>quantity</u>	<u>total_price</u>
1	101	201	2024-05-01	2	50.00
2	102	202	2024-05-02	1	25.00
3	103	201	2024-05-02	3	75.00
4	101	203	2024-05-03	2	60.00
5	104	204	2024-05-04	1	30.00
6	102	205	2024-05-05	4	100.00
7	101	201	2024-05-06	2	50.00
8	105	206	2024-05-06	3	90.00
9	103	207	2024-05-07	1	40.00
10	106	208	2024-05-08	2	55.00

- 2) Design a set of SQL queries representing various data retrieval operations, including simple lookups, range queries.
- 3) Create multiple index types (e.g., B-tree, Hash, Bitmap) on selected columns of the dataset.
- 5) Normal Query with out INDEX to measure the execution time of queries
- 6) Query with BTREE INDEXING to measure the execution time of queries
- 7) Normal Query with out INDEX to measure the execution time of queries
- 8) Query with BTREE INDEXING to measure the execution time of queries

Post-Lab:

1) What was the purpose of creating different index types on the dataset?

2) Describe the process you followed to create the different index types. Which indexing techniques did you use?

3) How did you measure the query performance before and after creating indexes?

4) What were the key metrics used to analyze the impact of indexes on query performance?

5) Did you observe any significant differences in query performance after implementing indexes? If so, elaborate on the changes.

6) Discuss any challenges or limitations encountered during the experiment.

7) How do you think the choice of index type influenced query performance?

8) Can you explain any trade-offs involved in selecting different index types?

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u> 	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 09

9. Query Optimization

Aim:

The aim of this experiment is to understand and apply various query optimization strategies to enhance the performance of database queries.

Description:

This involves analyzing the execution plans of queries, identifying bottlenecks, and implementing optimization techniques such as indexing, query rewriting, and the use of efficient algorithms. By the end of the experiment, students should be able to:

- Understand the importance of query optimization in database management systems (DBMS) and how it impacts overall system performance.
- Analyze query execution plans to identify performance bottlenecks and inefficient operations.
- Implement indexing strategies to improve the speed of data retrieval operations.
- Apply query rewriting techniques to transform queries into more efficient forms without changing their semantics.
- Evaluate the performance improvements achieved through various optimization strategies by comparing query execution times before and after optimization.

Pre-Requisites: PostgreSQL, TerraER Tool, Windows/ Ubuntu/CentOS/Debian, DBMS Concepts.

Pre-Lab:

1) What is query optimization in the context of a DBMS?

2) Why is query optimization important for database performance?

3) Explain the role of the query optimizer in a DBMS.

4) What is an execution plan in the context of query optimization?

5) What is query rewriting, and how can it improve query performance?

6) What is the purpose of analyzing query execution plans?

In-Lab:

1) Query optimization strategies for improving query performance.

a) Analyzing Execution Plans

Objective: Understand how to generate and analyze execution plans to identify performance bottlenecks.

Steps:

1. Create and Populate Tables

2. Write Complex SQL Queries

3. Generate Execution Plans.

Questions:

1) Create tables for employee and department with appropriate key constraints?

2) Write Complex SQL Queries Involving Joins and Subqueries.

3) Use the EXPLAIN command to generate execution plans for each query.

b). Identifying Performance Bottlenecks

Objective: Identify performance bottlenecks in SQL queries and understand their impact on query performance.

Questions:

1) Create tables for customers and orders with appropriate key constraints?

2) To write SQL queries with potential performance issues, identify these issues using the EXPLAIN command, and understand their impact on query performance.

c). Query Optimization

Query Optimization in a Sales Database

Objective: To optimize a query in a sales database by identifying and resolving performance bottlenecks.

Apply Optimization Techniques:

- Add indexes on frequently queried columns.
- Rewrite complex joins and subqueries.

Use EXISTS instead of IN for subqueries when appropriate

1) Create tables for customers and orders with appropriate key constraints?

2) Write a query to retrieve sales data for products in a specific category

3) Create appropriate indexes to optimize the query.

4) Measure and compare the performance before and after optimization using execution plans and query execution times.

5) Optimize SQL queries based on identified performance bottlenecks to improve query efficiency.

Viva-Voce Questions:

1) What is the significance of using indexes in SQL queries, and how do they improve query performance?

2) Explain how the EXPLAIN statement can be used to identify performance bottlenecks in SQL queries.

3) What are composite indexes, and when should they be used in query optimization?

4) How can query performance be affected by the use of functions in the WHERE clause, and what are the alternatives to mitigate this impact?

5) Explain the importance of query execution time measurement and the methods to measure and compare query performance before and after optimization.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u> 	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 10

10. NoSQL Experiment

Aim:

To understand the basics of NoSQL databases, design a schema, model data, and perform CRUD (Create, Read, Update, Delete) operations using a NoSQL database.

Description:

NoSQL databases are a type of database management system that diverges from the traditional relational database model used in SQL databases. They provide flexible data models and are designed to handle large volumes of data and high concurrency. NoSQL databases are commonly used in modern web applications, big data analytics, and real-time data processing scenarios.

Pre-Requisites:

- Choose a NoSQL database to work with (e.g., MongoDB, Cassandra, Redis)
- Install the database software on your local machine or use a cloud-based service.
- Set up a development environment with appropriate drivers or libraries for your chosen NoSQL database.
- Access the database through a command-line interface or a graphical user interface.

Pre-Lab:

1) What distinguishes NoSQL databases from traditional SQL databases, and why are they increasingly used in modern application development?

2) Can you name at least two real-world scenarios where NoSQL databases might outperform SQL databases, and explain why?

3) Describe the concept of schema flexibility in NoSQL databases and discuss its implications for database design and development?

4) Name and briefly explain two types of NoSQL databases, highlighting their respective strengths and ideal use cases?

2. Key-Value Store (e.g., Redis):

5) How do NoSQL databases typically handle scalability and high availability, and why are these features important for modern applications?

6) Compare and contrast the querying capabilities of NoSQL databases with those of SQL databases, considering factors such as query language and indexing.

In-Lab:

E-commerce Platform

You are developing an e-commerce platform where users can browse products, add items to their cart, and place orders. Design a MongoDB schema and implement CRUD operations to manage products, user carts, and orders? Design appropriate schemas and implementing CRUD operations to below mentioned questions and meet the needs of each use case.

1) Develop a query to store a product information including name, description, price, and stock quantity?

2) Write a query to create accounts with usernames, email addresses, and passwords?

3) Develop a query to perform operation on Users can add/remove products from their shopping carts?

4) Write a query where Users can place orders containing items from their cart?

5) Develop a query for to Delete products, user carts, and orders.?

Viva-Voice Questions (In-Lab):

1) What are the advantages of using NoSQL databases over traditional SQL databases?

2) Can you explain the concept of schema flexibility in NoSQL databases?

3) How do NoSQL databases handle scalability and high availability?

4) What are some common types of NoSQL databases, and how do they differ from each other?

5) In what scenarios would you recommend using a NoSQL database instead of a SQL database?

Post Lab:

You are developing a task tracking application for managing project tasks and deadlines. Design a MongoDB schema and implement CRUD operations to manage projects, tasks, and user assignments. Design appropriate schemas and implementing CRUD operations to below mentioned questions and meet the needs of each use case.

1) Develop a query where Users can create accounts with usernames, email addresses, and passwords?

2) Write a query to Users can create projects and assign tasks to team members.

3) Create a query to find Tasks have details such as title, description, due date, and status (e.g., pending, in progress, completed)?

4) Write a query to Update task details and project assignments?

5) Write a query to Perform operations to Delete tasks and projects.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 11

11. Big Data Basics(Use of Big Data Tool or Frame work for Basic Data Analysis & Transformation)

Pre lab:

- 1) What is Big Data Analytics?
- 2) What is Data Bricks?
- 3) What is SQL Analytics on Big Data?
- 4) What is Data Transformation?

5) Differentiate HADOOP and Apache Spark?

IN LAB:

Perform following Analytics on Apache Spark Big Data Environment Using Data Bricks.

sale_id	product_id	store_id	sale_date	quantity_sold	sale_amount
1	101	1	01-01-2024	10	100
2	102	1	02-01-2024	5	50
3	103	2	03-01-2024	20	200
4	101	2	04-01-2024	15	150
5	102	3	05-01-2024	25	250

- 1) Create the above Table and insert the values
- 2) Write SQL query to find out Total sales.
- 3) Write SQL Query to find Average Sales Amount per Transaction: Calculate the average sales amount per transaction.
- 4) Write SQL Query to find Total Quantity Sold: Calculate the total quantity sold

- 5) Write SQL Query to find Total Sales Amount by Product: Calculate the total sales amount grouped by product.

- 6) Write SQL Query to find Total Sales Amount by Store: Calculate the total sales amount grouped by store.

- 7) Write SQL Query to find Total Sales Amount by Date: Calculate the total sales amount grouped by sale date.

- 8) Write SQL Query to find Top 3 Stores by Sales Amount: Get the top 3 stores by sales amount

- 9) Write SQL Query to find Total Sales Amount in a Specific Date Range: Calculate the total sales amount within a specific date range.

Viva- Questions:

- 1) What is Databricks and how does it relate to big data analytics?
- 2) How does Databricks handle large-scale data processing?
- 3) What are the advantages of using Databricks for data analytics compared to traditional data processing frameworks?

4) Explain the role of Delta Lake in Databricks.

5) How can Databricks support machine learning workflows?

Post Lab:

employee_id	name	department	salary	hire_date	city
1	John Doe	Engineering	90000	15-01-2018	New York
2	Jane Smith	Marketing	80000	20-03-2019	Los Angeles
3	Sam Brown	Engineering	95000	10-06-2017	New York
4	Lisa White	HR	75000	25-11-2020	Chicago
5	Paul Green	Engineering	85000	30-05-2016	San Francisco
6	Nina Black	Marketing	78000	19-07-2018	Los Angeles

1) Create the above table and insert values Accordingly

2) List all employees in the Engineering department.

3) Calculate the average salary by department.

4) Find employees hired after January 1, 2018.

5) Count the number of employees in each city.

6) List the top 3 highest paid employees.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---

Lab Session 12

12. Implement master-slave/master-master replication between two or more database instances

Pre lab:

1. Difference between replication and sharding in mongodb.
2. How many types of sharding exist in MongoDB?
3. Explain mongodb replication consistency.

Inlab:

1. Set up master-slave/master-master replication between two or more database instances.
2. Create a newdb on primary and check if that got replicated on first and second slave.

Viva

1. How is data stored in *collections* in *MongoDB*?
2. What is the default *port* on which *MongoDB* listens?
3. How does *MongoDB* provide high availability and disaster recovery?
4. Describe how to read data from a *MongoDB collection*.

Post lab:

1. create the necessary folders and files to create a new instance in MongoDB.
2. To distribute the load and choose the fastest route for the desired query, as well as to provide an interface between the client application and the cluster.

Students Signature

(For Evaluator's use only)

<u>Comment of the Evaluator (if Any)</u>	<u>Evaluator's Observation</u> Marks Secured: _____ out of _____ Full Name of the Evaluator: Signature of the Evaluator Date of Evaluation:
--	---